

Dataset collection and process

Dataset Preparation

The dataset used for the Vehicle Classification for Parking Management project was sourced from the publicly available PKLot dataset on Kaggle. The preparation process involved several key steps to ensure it was clean, consistent, and ready for use in machine learning models.

Download and Organization

- The dataset was downloaded from Kaggle and extracted into a structured directory with subfolders for each parking lot location.
- Each subfolder contained labelled images under varying environmental conditions.

Cleaning and Preprocessing

- All unreadable or corrupted image files were removed.
- Images were resized to a uniform resolution of 224x224 pixels to ensure consistency across the dataset.
- Filenames were standardized to indicate the vehicle type and parking status (e.g., `car_occupied_001.jpg`).
- Duplicate images were detected using hash-based comparison and removed to reduce noise in model training.

Hosting and Accessibility

- The prepared dataset and annotation file were uploaded to a dedicated branch in the project's GitHub repository, making it accessible to all team members for collaboration and integration.

How Others Can Use the Dataset

- To use this dataset for training and experimentation, users should begin by cloning the GitHub repository that hosts the prepared dataset. This can be done using standard Git commands. Once cloned, users should navigate to the `data/` directory where the cleaned images and the associated annotation file (`vehicle_data_labels.csv`) are located. This directory contains all necessary image data structured for easy access and integration.
- Next, users can load the dataset using Python. By employing libraries such as pandas, users can read the CSV file to obtain structured labels that pair with each image. This step is essential for any downstream machine learning processes, including model training and validation.
- The dataset has been prepared in a format that aligns with standard machine learning workflows, particularly those involving convolutional neural networks (CNNs). This

structured approach enables users to plug the dataset directly into their pipelines for tasks such as image classification, segmentation.

- Overall, this setup ensures that the dataset is not only reusable but also promotes consistency in experimentation and collaborative development across the project team.

Dataset Dictionary

The following table describes the structure of the dataset annotation file (vehicle_data_labels.csv) used to support model training and evaluation:

Column Name	Data Type	Description
image_path	String	The relative file path to the corresponding image file.
vehicle_type	String	The classification label for the vehicle: 'car' or 'truck'.
slot_status	String	Indicates whether the parking slot is 'occupied' or 'vacant'.